

# National Cadastral Document QR & Barcode Verification Master Plan

## Dual-Factor Physical Sticker Seal & Tamper-Evident Digital Verification Architecture

**Executive Summary:** In land administration, physical deeds (7/12 Satbara, Sale Deeds, Property Cards) are vulnerable to forgery, illicit tampering, and duplicate bank mortgages. The ILRDVS QR Verification System establishes a cryptographic bridge between physical paper documents and the state's cloud registry using **High-Density 2D QR Codes**, **Code 128 Barcodes**, and **Secret Security PINs (SEC-XXXX-YYYY)**. This master brief details the operational protocol, verification tiers, security guarantees, and deployment workflows.

### 1. The Core Problem & Dual-Factor Solution

A purely digital database cannot prevent someone from presenting an altered paper photocopy in rural areas. Conversely, an unlinked paper stamp can be duplicated using desktop printers. ILRDVS integrates **Dual-Factor Document Authentication**:

| Factor 1: Digital Anchor (Public ID)                                                                                                                                                                                                                                                                             | Factor 2: Physical Seal (Secret PIN)                                                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li><b>Unique Document ID:</b> Assigned per record (e.g. DOC-MH-2026-1001).</li><li><b>Registry Match:</b> Connects to owner, survey/gat number, khata, area, and village.</li><li><b>SHA-256 Checksum:</b> Cryptographic fingerprint of the original archival scan.</li></ul> | <ul style="list-style-type: none"><li><b>Secret Security PIN:</b> 8-character cryptographic token (e.g. SEC-DA2C-7DA3).</li><li><b>Physical Sticker Seal:</b> Printed on 80mm adhesive sticker and affixed to paper deed.</li><li><b>Proof of Genuine Document:</b> Confirms physical paper deed matches cloud archive.</li></ul> |

### 2. The 3-Tier Verification Protocol

Every verification query processed by the portal (via camera scan, barcode gun, or manual search) is evaluated into one of three security tiers:

| Security Tier                    | Trigger Condition                                                      | System Verification Response                                                                              | Operational Action                                  |
|----------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| <b>Tier 1</b><br>■ Digital Only  | User enters Document ID without Secret PIN.                            | Digital archive record is confirmed valid. System prompts for physical sticker PIN.                       | Verify physical paper deed by entering sticker PIN. |
| <b>Tier 2</b><br>■ Authenticated | Phone camera scans sticker QR code OR user enters matching Secret PIN. | <b>Physical Sticker Authenticated:</b> Secret PIN matches cryptographic state registry. Full green badge. | Paper document accepted as legally authentic.       |
| <b>Tier 3</b><br>■ Tamper Alert  | Supplied Secret PIN does NOT match the document record.                | <b>Critical Tamper Warning:</b> Forged physical sticker or altered paper photocopy detected.              | Immediate rejection. Escalate to Sub-Registrar.     |

### 3. The Cryptographic Secret Security PIN Engine

The Secret PIN is generated using non-linear multiplicative bit-mixing. It operates deterministically from the document's unique ID and a state salt seed:

#### Mathematical Formula:

Input = DocumentID + ':ILRDVS-MAHA-SEAL-2026'  
 $h1 = \text{imul}(h1 \wedge \text{ch}, 2654435761)$ ;  $h2 = \text{imul}(h2 \wedge \text{ch}, 1597334677)$ ;  
Token = 'SEC-' + Hex(h1)[0..4] + '-' + Hex(h2)[0..4] → **SEC-DA2C-7DA3**

- Avalanche Diffusion:** Altering 1 character in Document ID completely randomizes the resulting PIN.
- Brute-Force Resistance:**  $16^8 = 4,294,967,296$  possible combinations. Mathematically impossible to guess.
- API Privacy Masking:** Online public API responses mask the PIN (SEC-\*\*\*\*-7DA3) to prevent scraping.
- Universal Phone Compatibility:** Standard camera scan decodes the URL and PIN without requiring special apps.

## 4. Physical Adhesive Sticker Specifications

The physical sticker is calibrated to standard thermal label printer dimensions (80mm x 80mm / 3in x 3in) and is affixed to the top-right corner of Page 1 of the physical paper deed:

| Element                | Specification                      | Purpose & Operational Function                                                             |
|------------------------|------------------------------------|--------------------------------------------------------------------------------------------|
| Physical Dimensions    | 80mm x 80mm Square Label           | Fits thermal label printers (Zebra, Brother, TVS) and standard A4 adhesive sheets.         |
| State Authority Header | Govt of Maharashtra • Revenue Dept | Official state emblem indicating statutory land revenue registry seal.                     |
| 2D QR Code             | Error Correction Level M (280px)   | Encodes direct verification URL with embedded secret PIN for 1-tap phone scanning.         |
| 1D Barcode             | Code 128 (High Density)            | Scannable by handheld optical barcode guns in taluka sub-registrar record rooms.           |
| Secret PIN Badge       | High-contrast Gold Monospace       | Visible PIN for manual two-factor citizen / banking verification.                          |
| Affix Instruction      | Top-Right Page 1 Deed Placement    | Ensures immediate visibility upon file inspection and protects against sheet substitution. |

## 5. Stakeholder Operational Workflows

- A. Citizen / Land Buyer:** Seller presents paper 7/12 extract → Buyer points smartphone camera at sticker → Browser opens immediately with ■ Authenticated Badge, real owner name, survey/gat number, and area.
- B. Bank Loan Officer:** Farmer submits deed for agricultural mortgage → Officer enters ID & Secret PIN on workstation → Confirms deed is genuine and checks latest encumbrance/mutation history before loan approval.
- C. Sub-Registrar / Revenue Court:** During deed conveyance or boundary dispute → Officer scans barcode gun → Digital twin archival scan is loaded side-by-side to verify zero handwritten fraudulent alterations exist.

## 6. Production API & Cadastral Workstation Reference

- **Public Unauthenticated Verification API:** GET /api/documents/public-verify/:query?sec=:pin
- **Public Verification Portal Web Route:** /verify-document?id=:docId&sec=:pin
- **Cadastral Dual-Pane Inspection Workstation Actions:**
  1. **QR Sticker:** Instant one-click download of 300 DPI high-resolution sticker PNG image.
  2. **QR Seal:** Modal with physical sticker preview, direct 80mm thermal printing, and URL test.
  3. **Download PDF:** Official digital certificate export with statutory notice and audit seal.
- **Immutable Audit Trail:** Every verification attempt (IP address, timestamp, result tier) is logged in PostgreSQL.

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